Deep Learning Framework using MCLAHE with Pretrained CNNs and Sequence based Models for Histopathological Colon Cancer Classification

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***Abstract*— Colon cancer is one of the leading causes of cancer-related deaths worldwide, and early detection is vital for improving patient outcomes. This paper presents a deep learning-based framework for automated colon cancer classification using microscopic histopathological images. The primary objective is to systematically compare and evaluate multiple architectures to identify the most reliable diagnostic model. A dataset of 10,000 images, including 2,000 for independent testing, was preprocessed using Multiscale CLAHE (MCLAHE) to enhance structural details. Features were extracted with three pretrained CNNs EfficientNetB3, VGG19, and ResNet152 and classified using Neural Networks, LSTM, and BiLSTM, resulting in nine configurations. Performance was measured with accuracy, F1-score, MCC, Youden's Index, G-Mean, and Cohen's Kappa. Results demonstrate that CNN-BiLSTM hybrids achieved superior performance, with EfficientNetB3 + BiLSTM reaching 99.80% accuracy. The study contributes a robust and reproducible framework that supports precise, automated colon cancer diagnosis.**

Keywords—BiLSTM, Colon Cancer, Deep Learning, EfficientNet-B3, Feature Extraction, Histopathological Images, LSTM, MCLAHE, Neural Network, Preprocessing, ResNet-152, VGG16.

I. Introduction

Colon cancer refers to a cancerous tumor that originates in the colon, an essential component of the digestive system and the foremost portion of the large intestine. In 2012, approximately 1.4 million individuals were diagnosed, and in 2017, the disease resulted in almost 50,260 deaths [1]. Across the globe, colon cancer is the third most frequently diagnosed cancer in men and the second in women. Its prevalence differs by geographic location, age, and sex and is projected to rise by 80% as of 2035 [2]. Colon cancer goes through four stages. Stage I involves only the inner lining of the colon. During Stage II, tumors develop in the wall but not in lymph nodes or other tissues. Stage III involves adjacent lymph nodes, while Stage IV describes metastasis to distant organs such as the lungs [3]. Timely detection plays a crucial role in improving treatment outcomes. Today, CT, MRI, colonoscopy, histopathology, and biomarker tests are being used for diagnosis [4]. Recently, machine learning and deep learning have enhanced diagnostic accuracy by automatically extracting complex features from histopathological images, outperforming traditional techniques.

This paper is organized as follows. Section II describes the dataset and preprocessing methods. Section III outlines the proposed methodology, including feature extraction and model architectures. Section IV explains the experimental setup and evaluation metrics. Section V presents the results and provides a comparative discussion. Finally, Section VI concludes the paper and highlights potential future research directions.

II. Literature Survey

CLAHE has proven effective in boosting local contrast in medical images with minimal noise enhancement, making it a valuable tool for diagnostic imaging across multiple biomedical applications [5]. Conventional classifiers like SVM, Random Forest, and XG Boost, and deep learning-based models like CNNs, have demonstrated potential in cancer tissue classification [6]. CNN-based models, especially, have reported high accuracy in colon adenocarcinoma diagnosis, with even higher performance than traditional approaches. Ensemble models also enhance robustness and reliability [7]. Sophisticated pretrained CNN models such as EfficientNet and ResNet have proven to be better in the diagnosis of colorectal cancer for early detection and clinical decision-making [8]. In order to model spatial dependencies in histopathological images, new strategies combine CNNs with sequence models like LSTM and BiLSTM, transforming extracted features into sequence forms that maintain tissue relationships [9]. Additionally, weakly supervised learning methods like CLAM and modified CNNs employ the attention mechanism for localizing diagnostic areas without needing extensive annotations, promoting interpretability and efficiency [10]. Transfer learning with models like EfficientNetB0 and ResNet50 has also enhanced sensitivity and minimized training time in cancer classification tasks [11]. A Study applied GoogLeNet and VGG-19 to extract features from LC25000 histology images, followed by PCA for dimensionality reduction and classification using ANN. Fusion with handcrafted features further improved the model's sensitivity, precision, and specificity for lung and

colon cancer detection [12]. Existing CNN-based histopathology methods often struggle with artifacts, poor staining, and small tumor regions, resulting in misclassification. Most approaches rely on a single CNN backbone or a limited set of classifiers, restricting their effectiveness. We address this by combining MCLAHE preprocessing with EfficientNet-B3, VGG19, ResNet-152 for feature extraction and NN, LSTM, and BiLSTM for classification. This hybrid framework enhances feature quality and contextual learning, improving accuracy and robustness in colon cancer classification.

III. Materials and Methods

3.1 Collection of Dataset and Preprocessing

In this work, a colon cancer histopathological publicly available image dataset was used with resolution of 224×224 pixels [13]. All tissue samples were stained using Hematoxylin and Eosin (H&E), comprising a total of 10,000 microscopic images. These images are equally distributed across two classes: normal and cancerous colon tissue. For the purpose of ensuring strong model testing, the data was separated into distinct subsets for training and testing purposes, where 8,000 images (80%) were used for training purposes and 2,000 images (20%) were reserved for testing. The classes were represented equally within both subsets to ensure class balance. All images underwent pre-processing via Multiscale Contrast Limited Adaptive Histogram Equalisation (MCLAHE). CLAHE is a method that helps enhance local contrast by implementing histogram equalization on small tiles rather than the entire image. Let the input grey level image be $I(x, y) \in [0, L-1]$. Divide the image into M x N tiles, and for each tile T_k, calculate a histogram H_k. To avoid noise amplification as shown in the equation (1) histogram values are clipped at β:

$$H_k^{clipped}(i) = \min(H_k(i), \beta) \quad (1)$$

The clipped pixels are redistributed uniformly as shown in the equation (2)

$$H_k^{final}(i) = H_k^{clipped}(i) + \frac{C}{L} \quad (2)$$

After that, the CDF is calculated, and the pixel values are mapped as shown in the equation (3):

$$I'_k(x, y) = (L-1).\,CDF_k(i) \quad (3)$$

To reduce block artifacts, bilinear interpolation is applied between adjacent tiles [14]. We used a tile size of 8×8 and a clip limit of 2.0, chosen to enhance local contrast while minimizing noise and block artifacts. These values were found to provide the best balance between visual clarity and classification performance.

3.2 Methodology

After preprocessing, images were resized according to the input requirements of each pretrained transfer learning model used for feature extraction. These enhanced and normalized images were then passed to pretrained convolutional neural networks to extract meaningful feature representation. Then Hyperparameters (learning rate = $1e^{-4}$, batch size = 32, dropout = 0.5, LSTM/Bi-LSTM units = 64) were chosen through manual tuning based on validation accuracy and loss curves. Models were trained for 20 epochs using the Adam optimizer with binary cross-entropy loss, with dropout regularization to reduce overfitting. The overall methodology of this research work is described in Fig 1.

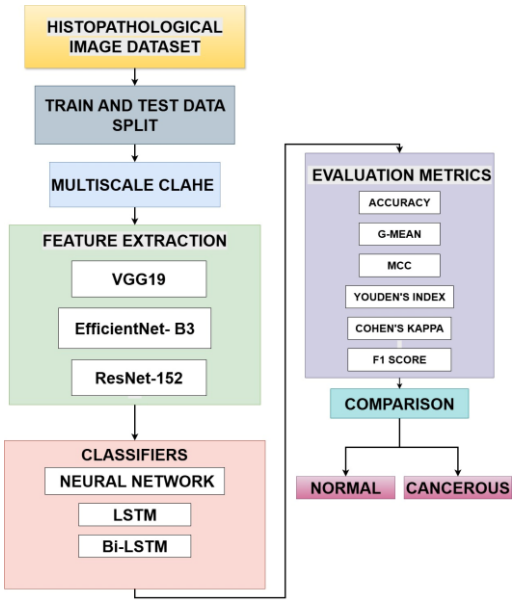


Fig 1. Methodology for colon cancer detection

IV. Transfer Learning Architecture For Feature Extraction

This study employed three pretrained CNN architectures such as EfficientNet-B3, VGG-19 and ResNet-152 for extracting high-level and discriminative features from the images enhanced using the MCLAHE technique. The objective was to enhance the quality of feature representation and subsequently improve the accuracy and effectiveness of the classification process.

4.1 EfficientNet-B3:

As demonstrated in fig.2 Input images were set to 300×300 pixels and passed through the network to generate a 10×10×1536 feature map. A GlobalAveragePooling2D layer was used to reduce the spatial dimensions, yielding a 1536-dimensional feature vector per image. These vectors were then passed into NN, LSTM, and BiLSTM classifiers.

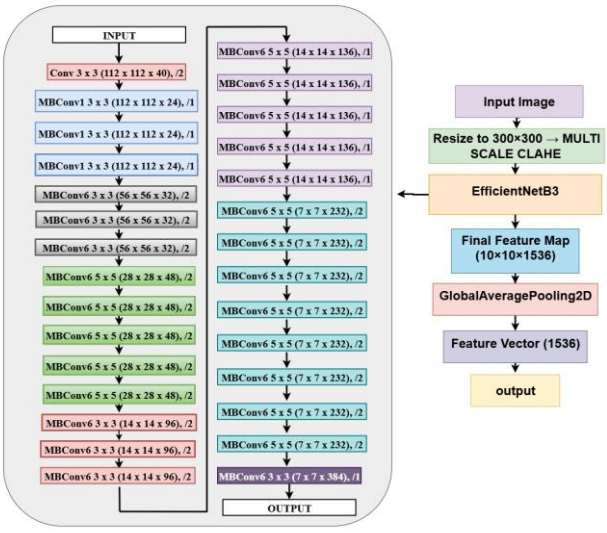


Fig 2. EfficientNet-B3 Architecture

4.2 VGG-19:

As shown in fig.3 All input images were set to 224×224 pixels and passed through the network to obtain a final convolutional feature map of size 7×7×512. A GlobalAveragePooling2D layer was then applied to compress this feature map into a 512-dimensional feature vector per image.

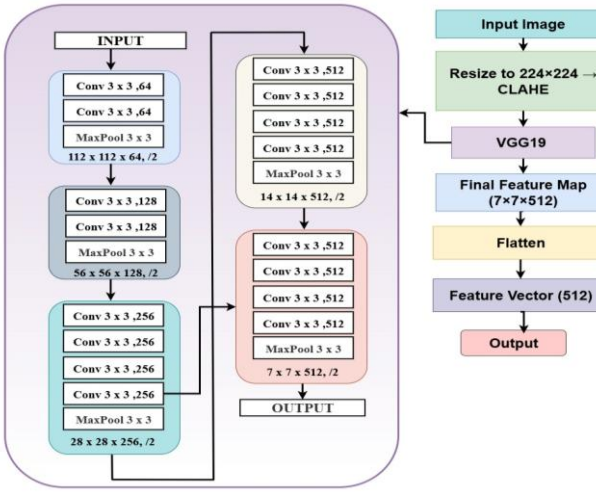


Fig 3. VGG-19 Architecture

4.3 ReseNet-152:

As depicted in Fig.4 Input images were set to 224×224 pixels as needed by ResNet152 input specifications, and then passed through the network to generate 7×7×2048 feature map. A GlobalAveragePooling2D layer was used to reduce the spatial dimensions, yielding a 2048-dimensional feature vector per image.

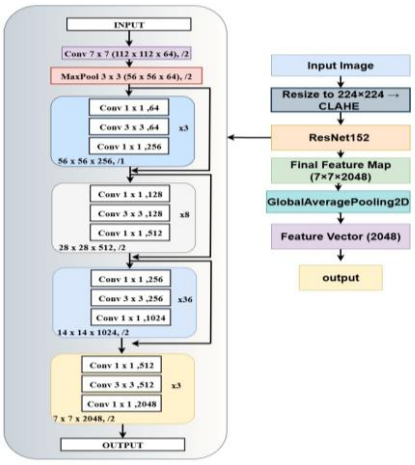


Fig 4. ResetNet-152 Architecture

V. Deep Learning Based Classifiers

To capture deeper relationships within CNN-extracted features, LSTM and BiLSTM models were applied after feature extraction as depicted in Fig 1. Though the features are not time-series, sequential models can learn inter-feature dependencies and spatial progression patterns in tissue structures.

5.1 Neural Network:

After extracting spatial features from each image using these CNNs, the feature values are flattened into one-dimensional vectors which is to be compatible with a fully connected architecture. These feature vectors served as inputs to the dense network.

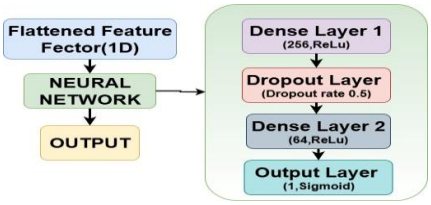


Fig 5. Neural Network Architecture

As depicted in Fig.5 The architecture includes a Dense layer with 256 units and ReLU activation, followed by a Dropout layer with a rate of 0.5 to mitigate overfitting. This is succeeded by another Dense layer comprising 64 units with ReLU activation. Finally, a Dense output layer with sigmoid activation is used to perform binary classification.

5.2 LSTM:

As depicted in Fig.6 A memory cell and three gates which includes the forget, input, and output gates that regulate information flow make up each LSTM unit. Which portion of the prior cell state should be discarded is determined by the forget gate F_t. The input gate I_t regulates the amount of new data that should be added to the cell. The output gate decides O_t if the memory cell should contribute to the output at the current time step as shown in the equations (4), (5) and (6) [15].

$$F_t = \sigma(W_f.[h_{t-1}, x_t] + B_f) \quad (4)$$

$$I_t = \sigma(W_i.[h_{t-1}, x_t] + B_i) \quad (5)$$

$$O_t = \sigma(W_o.[h_{t-1}, x_t] + B_o) \quad (6)$$

In our study, pretrained CNNs features were reshaped into a 2D sequence and passed into LSTM network. To produce the final prediction, the final hidden state was run through fully connected layers. By integrating LSTM with CNN-extracted features, our approach leverages both spatial representation and sequential model to improve the classification of colon cancer histology image.

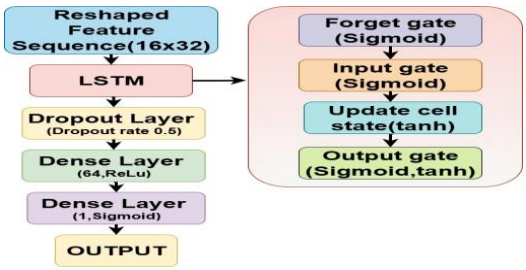


Fig 6. LSTM Architecture

5.3 BiLSTM:

To further boost the sequential learning ability BiLSTM was utilized, which reads input sequences both in forward and backward directions as depicted in Fig 7. This improves the network's pattern recognition ability from CNN-extracted features by allowing it to learn dependencies from past and future time steps.

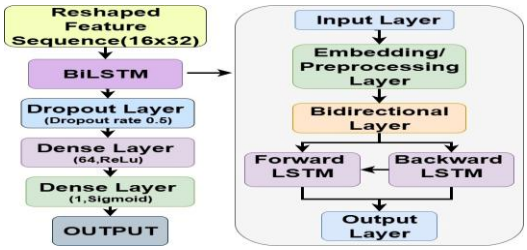


Fig 7. BiLSTM Architecture

In BiLSTM, for each time step t, two hidden states are generated: one from the forward pass $\overrightarrow{h_t}$ and another from the backward pass $\overleftarrow{h_t}$ .The final output at each step is the concatenation of these two as shown in the equation (7):

$$h_t = \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right] \quad (7)$$

This bidirectional representation was then passed through dense layers and sigmoid for final classification.

VI. Results And Discussion

This section presents a comprehensive analysis of the model performance in classifying histopathological images. The classifiers used in this study are a simple neural network, LSTM, and BiLSTM. Six performance metrics like Accuracy, G-Mean, MCC, Cohen's Kappa, F1-Score and Youden's Index were used to assess each combination's performance both quantitatively and qualitatively using equations (8), (9), (10) (11), (12), (13) and (14).

Accuracy: The percentage of all accurate forecasts, both positive and negative.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (8)$$

G-Mean (Geometric Mean): Balances performance on both classes, especially useful for imbalanced datasets.

$$\text{GMean} = \sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}} \quad (9)$$

Matthews Correlation Coefficient (MCC): A balanced metric ranging from -1 to +1, with 0 indicating random guessing even with imbalanced classes.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (10)$$

Cohen's Kappa (K) measures the level of agreement between predicted and actual outcomes, while adjusting for chance agreement.

$$K = \frac{P_0 - P_e}{1 - P_e} \quad (11)$$

F1 Score combines precision and recall into a single metric and is especially valuable when dealing with class imbalance.

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precison} + \text{Recall}} \quad (12)$$

Youden's Index (J): Measures the effectiveness of a binary classifier, ranging from -1 to 1.

$$\text{Youden's Index} = \frac{TP}{TP+FN} + \frac{TN}{TN+FP} - 1 \quad (13)$$

Fig. 8 shows the training and validation accuracy and loss of EfficientNet-B3 + NN over 20 epochs. The model achieves steadily increasing accuracy with decreasing loss, indicating effective learning and good generalization without significant overfitting.

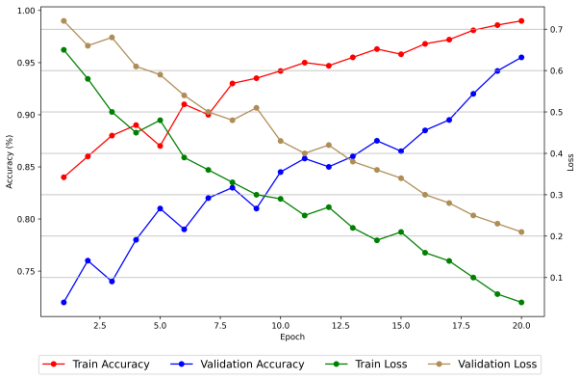


Fig.8 EfficientNet-B3 + NN accuracy and loss over Epochs

In EfficientNet-B3 + LSTM as depicted in Fig.9 we observed faster convergence with the validation accuracy surpassing 98% by the 18th epoch. The training and validation curves were tightly coupled, demonstrating a well-generalized model with minimal overfitting.

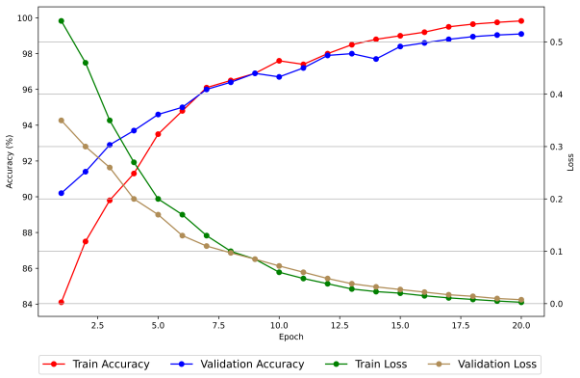


Fig.9 EfficientNet-B3 + LSTM accuracy and loss over Epochs

Fig.10 Illustrates that the EfficientNet-B3+Bi-LSTM configuration achieved almost perfect overlap between training and validation curves. It reached a validation accuracy of 99.8% and near-zero loss by epoch 20, indicating stable learning and exceptional model generalization.

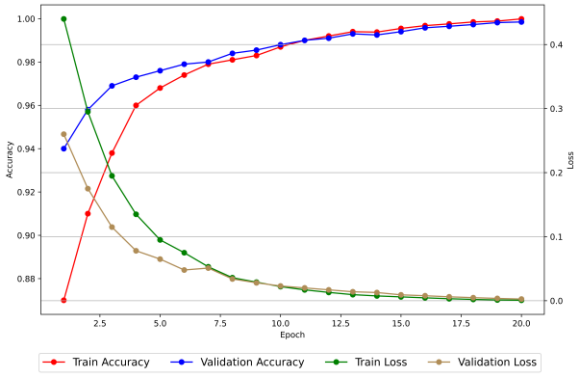


Fig.10 EfficientNet-B3 + BiLSTM accuracy and loss over Epochs

For ResNet152 + NN as shown in Fig .11 validation accuracy plateaued at 88% initially but gradually rose to 97% after epoch 12. This suggests that the model began learning complex patterns only in the latter epochs.

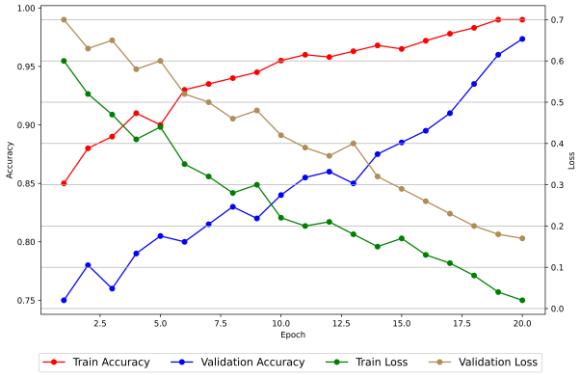


Fig.11 ResNet152+ NN accuracy and loss over Epochs

In ResNet152 + LSTM as indicated in Fig.12 validation accuracy of 98% was achieved as early as the 6th epoch. The loss curve flattened below 0.02, highlighting rapid and stable convergence.

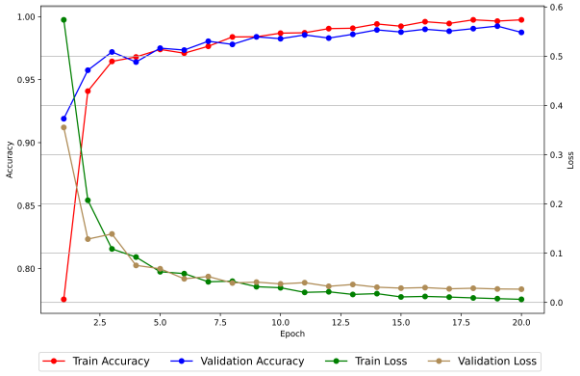


Fig.12 ResNet152+ LSTM accuracy and loss over Epochs

The ResNet152+Bi-LSTM model depicted in Fig.13 achieved 99% validation accuracy at the 4th epoch and maintained it with negligible loss, making it the quickest-converging model in this work.

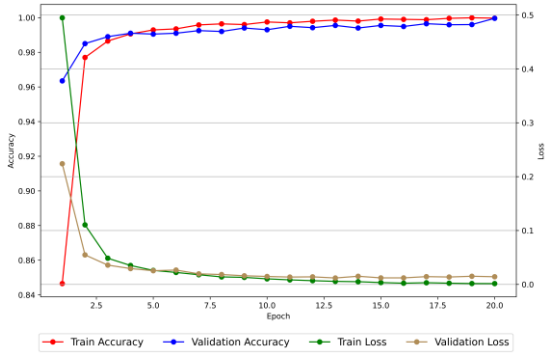


Fig.13 ResNet152+BiLSTM accuracy and loss over Epochs

With VGG19 + NN as displayed in Fig.14 validation accuracy increased from 75% to 97%, but a consistent gap with the training accuracy indicated overfitting beyond epoch 14.

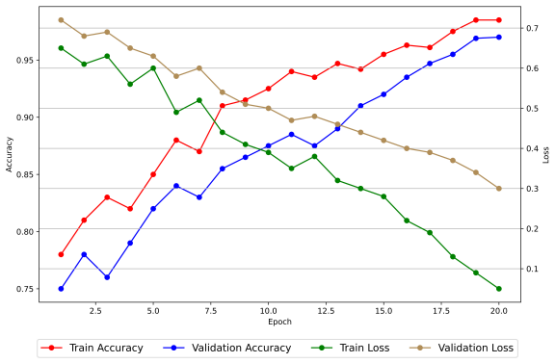


Fig.14 VGG19+ NN accuracy and loss over Epochs

Finally, the VGG19 + LSTM as depicted in Fig.15 and VGG19 + Bi-LSTM as depicted in Fig.16 showed significant improvements in generalization. The Bi-LSTM version achieved 99% validation accuracy, clearly benefiting from its ability to model bidirectional temporal dependencies

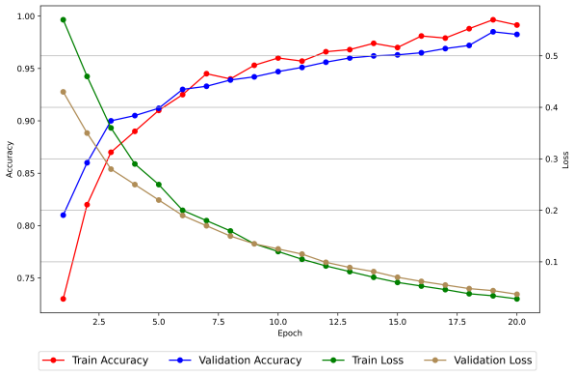


Fig.15 VGG19+ LSTM accuracy and loss over Epochs

These observations confirm that the inclusion of temporal sequence models, especially Bi-LSTM, enables the model to effectively identify and learn detailed patterns from image features.

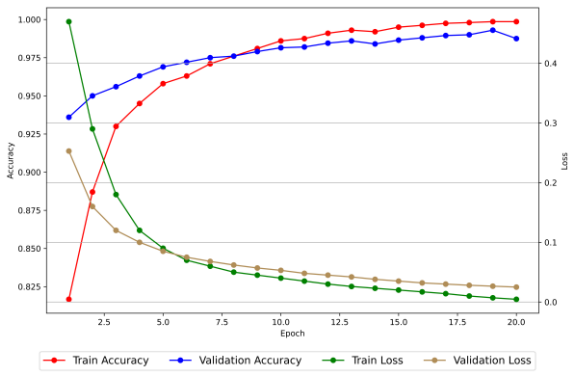


Fig.16 VGG19+ BiLSTM accuracy and loss over Epochs

To validate model performance, six standard classification metrics were used. The table 1 shown below displays the comparison results of different transfer learning architectures along with the deep learning-based classifiers.

Achieving an accuracy rate of 99.80%, MCC of 0.996 and F1-score of 99.80% the EfficientNet-B3 + Bi-LSTM model performed the best, showing almost perfect correlation between predicted and actual labels. These metrics attest to the hybrid architecture's superior performance, which successfully blends temporal sequence modelling and spatial

feature extraction. Notably, the Bi-LSTM variants consistently outperformed both LSTM and NN-based classifiers across all three backbones. This confirms the advantage of bidirectional information flow in capturing dependencies in the extracted feature sequences, a critical aspect in medical imaging where structural patterns may not be strictly sequential. To validate that the observed performance differences are not due to chance, we applied McNemar's test between the best-performing EfficientNet-B3 + BiLSTM model and other competing models. The test indicated statistically significant improvements ($p < 0.05$), confirming the superiority of EfficientNet-B3 + BiLSTM.

The EfficientNet-B3 base model without recurrent units yielded the highest number of false positives (FP) indicating that while it was sensitive to detecting cancerous patterns, it often misclassified benign images. In contrast, the EfficientNet-B3 + Bi-LSTM model recorded only one false negative (FN) and seven false positives, highlighting the clinical value of this model minimizing false negatives is critical in cancer diagnostics, as these errors could result in missed detections. Error cases were visually inspected, and it was found that most misclassified tiles contained blurred cell boundaries, poor contrast, or staining artefacts, which complicate the feature extraction process. Future improvements could focus on enhancing the preprocessing pipeline to compensate for such artefacts. False negatives can delay diagnosis, and false positives may cause unnecessary tests; our model serves as a decision support tool to assist clinicians, improve accuracy, and streamline workflow.

Table:1 Comparison of Deep Learning Model Performances

Model	Accuracy (%)	Youden's Index (%)	Geometric-Mean (%)	MCC	Kappa	F1-Score (%)
VGG-19 + NN	94.00	0.87	93.51	0.87	0.87	92.40
VGG-19 + LSTM	97.00	0.93	96.70	0.93	0.94	96.21
VGG-19 + BiLSTM	98.75	0.97	98.64	0.97	0.97	98.43
ResNet-152+ NN	95.25	0.90	95.23	0.90	0.91	95.16
ResNet-152+ LSTM	97.90	0.95	97.90	0.95	0.96	97.88
ResNet-152+ BiLSTM	99.00	0.98	99.00	0.98	0.98	98.99
EfficietNet-B3+ NN	95.65	0.91	95.64	0.91	0.91	95.70
EfficietNet-B3+ LSTM	98.00	0.96	97.99	0.96	0.96	98.96
EfficietNetB3+ BiLSTM	99.80	0.99	99.80	0.99	1.00	99.80

VII. Conclusion

In this work, we proposed a deep learning pipeline that integrates multiscale-CLACHE preprocessing, advanced CNN-based feature extraction, and Bi-LSTM-based classification for histopathological image analysis. Among all tested configurations, the EfficientNet-B3 + Bi-LSTM model demonstrated exceptional performance, achieving 99.80% classification accuracy and 99.80% F1-score, while maintaining low error rates and robust generalization. Future research can include incorporating attention mechanisms to further improve feature attention within the Bi-LSTM. Investigating hybrid models such as vision transformers or graph networks could further enhance spatial knowledge. Alternative preprocessing methods like stain normalization and artifact-aware augmentation can further reduce misclassifications. Integrating explainability tools Grad-CAM, saliency maps with uncertainty estimation will strengthen clinical trust. Finally, optimizing the framework for lightweight deployment will support seamless integration into digital pathology workflows.

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